

API Documentation example Boreal

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Access to the API is provided to service accounts. The URL and credentials are provided on request. API access requires a REALM subscription.

Boreal tag traits example

Tag traits (tagID) are the unique identifier for a channel throughout the Brightlayer system. Therefore, the API works with tagIDs. The tagID for each channel can be retrieved from the ontology file. Examples:

- Name: "totalwater_in", tagID 10462371
- Name: "totalwater_pro", tagID 10462372

URL

The following URL (prefix) needs to be used to access the API on the EMEA Prod system:

<https://portal.machinery-monitoring.com>

Real-time API

POST <prefix>/api/v1/dashboard/devices/{deviceId}/realtime

This API will return the real-time data for a device. Only realtime data will be returned by this API.

Non-Normative Request Example

Request query parameter:

- deviceId : 9e153413-fcf1-2b11-0800-a887d221b600

Request body:

```
{
  "tagId": "string"
}
```

Request body example:

```
{
  "tagId": "10419813"
}
```

Non-Normative Response Example on Success

Response status code: 200

Response body:

```
{
  "v": {
    "unit": "",
    "value": "true"
  },
  "dt": "2025-07-17T08:00:44Z"
}
```

Response description:

- v - collection of the values returned from real-time query result:
 - unit - unit of the trait.
 - value - current value of datapoint/trait.
- dt - timestamp.

Time-series / Trends API

POST <prefix>/api/v1/dashboard/devices/timeseries

This will return the time-series data for a list of devices.

Non-Normative Request Example

Request body:

```
{
  "devices": [
    {
      "deviceId": "String",
      "tagTrait": "String"
    }
  ],
  "startDateTime": "DateTime (UTC)",
  "endDateTime": "DateTime (UTC)"
}
```

Example:

```
{
  "devices":[
    {
      "deviceId": "967071e5-83e1-2b11-0800-8c08ae746b00",
      "tagTrait": "10361718"
    }
  ],
  "startDateTime": "2025-07-15T00:00:00Z",
  "endDateTime": null
}
```

Request properties description:

- devices **(required)** - collection of the devices and tag/trait pairs for which the telemetry is queried:
 - deviceId **(required)** - id of the device.
 - tagTrait **(required)** - the tag trait for the device.
- startDateTime **(required)** - start time of the time range for time series query.
- endDateTime **(optional)** - end time of the time range for time series query.

Non-Normative Response Example on Success

Response status code: 200

Response body:

```
{
  "timeSeries": [
    {
      "tagTrait": "10361718",
      "deviceId": "9e153413-fcf1-2b11-0800-a887d221b600",
      "results": {
        "values": [
          {
            "v": {
              "unit": "kWh",
              "value": "123"
            },
            "dt": "2025-04-15T09:16:28Z"
          }
        ]
      }
    },
    {
      "tagTrait": "10415480",
      "deviceId": "9e153413-fcf1-2b11-0800-a887d221b600",
      "results": {
        "values": [
          {
            "v": {
              "unit": "kWh",
              "value": "456"
            },
            "dt": "2025-04-15T09:16:28Z"
          }
        ]
      }
    }
  ]
}
```

Response description:

- the response is a collection of the timeseries query result. Every element of that collection is differed by deviceId, tag.
- deviceId - GUID of the device.
- results - collection of the timeseries query results.
 - trait - trends data trait .
 - values - collection of timeseries values.
 - v - value of the trait.
 - dt - timestamp.

Command API

Remark: Currently no control channels are defined in the water_filtration ontology.

POST <prefix>/api/v1/dashboard/devices/{deviceId}/command

Update the device command sent from the cloud to the device. Device commands can only be sent to channels that are of the type “control”. The range of data, that can be sent to the device, is defined in the ontology.

Non-Normative Request Example

Request query parameter:

- deviceId : ba8ef5d8-b33d-42fd-85d0-416fa73345a0

Request body:

```
{
  "tagId": "String", // tag id for control channel
  "value" : "string" // data range needs to be defined in the ontology
}
```

Example:

1. Control command one

```
{
  "tagId": "10420585",
  "value" : "1"
}
```

2. Control command two

```
{
  "tagId": "10420586",
  "value" : "1"
}
```

Non-Normative Response Example

Response status code: 201 if value is set successfully.

Response status code: 404 if channel not found.

- tagIds(**required**) - collection of the tag ids for which data should be queried:
 - outletStatus(**required**) - tag id for the outlet status channel.
 - energyconsumptionactual(**required**) - tag id for the energy consumption actual channel.

Non-Normative Response Example on Success

Response status code: 200

Response body:

```
{
  "v": {
    "unit": "kWh",
    "value": "0.4"
  },
  "dt": "2025-07-17T10:37:35Z"
}
```

Response description:

- v - collection of the values returned from real-time query result:
 - unit - unit of the trait.
 - value - current value of datapoint/trait.
- dt - timestamp.

Devices API

GET <prefix>/api/v1/dashboard/organization/{organizationId}/devices

This API is designed for retrieving the device id and name mapping of devices that are present within an organization.

Non-Normative Request Example

Request query parameter:

- organizationId : 723af19b-95d6-43e0-b629-a18d1efe9225

Non-Normative Response Example on Success

Response status code: 200

Response body:

```
{
  "id": "723af19b-95d6-43e0-b629-a18d1efe9225",
  "devices": [
    {
      "id": "0639aba5-09a5-4671-a119-d6b684e81736",
      "name": "P0001-S0001-01"
    },
    {
      "id": "579b3195-fbd4-4cee-b857-99109445b9bb",
      "name": "P0002-S0001-01"
    },
    {
      "id": "e99b2311-a69a-40f0-92b0-5065be9164de",
      "name": "P0003-S0001-01"
    }
  ]
}
```

Response description:

- id - organization id.
- devices – collection of devices:
 - id – device id.
 - name – device name.

API to generate the token:

POST <prefix>/api/v1/auth/serviceaccount/token

Non-Normative Request Example:

Request body:

```
{  
  
  "serviceAccountId": "string",  
  "secret": "string"  
}
```

Non-Normative Response Example on Success

Response status code: 200

Response body:

```
{  
  "token":  
    "eyJraWQiOiJKaHQtQjRpWFpmRHVBUkF1Ykdra1NfM1o1SktkOVp3Q0pwM3laWlha"  
}
```